

# The pipe operator

Day 5 - Friday, May 22, 2026

**i** Today: Friday May 22

## Before class

- Read *Data Computing* §3.6 ([Click here](#)).

## Plan for today

- Quiz 1 (20 min)
- The pipe operator
- Chaining commands together
- In-class activity

## Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [Data Computing §3.6](#)

## Quiz 1

### Quiz 1 - instructions

**!** Closed-book, no AI, no electronics

- Put away phones and laptops.
- You may use a pen or pencil.
- 20 minutes.

### Topics covered:

- R as a calculator and basic syntax

- Data types and structures
- Objects and assignment
- Reading R code (functions, arguments, variables, constants)
- Packages: installing vs. loading
- Histograms in `ggplot2` (reading code, not writing from scratch)

When you finish, bring your quiz up to the front.

## Quiz debrief

- We'll go over common stumbling points on Monday after I've graded them.

## The pipe operator

### A problem we've already seen

Remember this code from Wednesday?

```
sqrt(mean(c(1, 4, 9, 16, 25)))
```

```
[1] 3.316625
```

This should be read from the **inside-out**:

1. First `c(1, 4, 9, 16, 25)` makes a vector.
2. Then `mean(...)` of that vector is 11.
3. Then `sqrt(11)` is  $\approx 3.317$ .

As the number of functions you nest grows, the harder and harder it gets to read.

### A real example

Suppose we want to:

1. Take the `penguins` data,
2. Keep only rows where `body_mass_g` is not missing,
3. Group by `species`,
4. Compute the mean body mass for each species.

Written with nested function calls:

```
summarize(  
  group_by(  
    filter(penguins, !is.na(body_mass_g)),  
    species  
  ),  
  avg_mass = mean(body_mass_g)  
)
```

You have to read it from the **inside out**, and the arguments get separated from their functions. This is not ideal or intuitive for humans.

## Enter the pipe

The **pipe operator** lets you read code left-to-right, top-to-bottom, which is the way you actually think about it.

```
penguins |>  
  filter(!is.na(body_mass_g)) |>  
  group_by(species) |>  
  summarize(avg_mass = mean(body_mass_g))
```

Read this as:

“Take **penguins**, **then** filter out missing values, **then** group by species, **then** summarize the average mass.”

That’s it. That’s the whole idea.

## What the pipe does

```
x |> f()
```

is **exactly the same** as

```
f(x)
```

The pipe takes whatever’s on the **left** and inserts it as the **first argument** to the function on the **right**.

```
# These do the same thing:  
sqrt(16)
```

```
[1] 4
```

```
16 |> sqrt()
```

```
[1] 4
```

### With multiple arguments

```
x |> f(arg2, arg3)
```

is the same as

```
f(x, arg2, arg3)
```

The pipe always fills the **first** argument; the rest go inside the parentheses as usual.

```
# Without the pipe:  
round(3.14159, digits = 2)
```

```
[1] 3.14
```

```
# With the pipe:  
3.14159 |> round(digits = 2)
```

```
[1] 3.14
```

## Chaining pipes

It is very useful to chain pipes together to do many steps in order.

```
c(1, 4, 9, 16, 25) |>
  mean() |>
  sqrt()
```

```
[1] 3.316625
```

Compare to the nested version:

```
sqrt(mean(c(1, 4, 9, 16, 25)))
```

```
[1] 3.316625
```

Here, we get the same answer. The piped version reads in the order you do the work.

### Style: one pipe per line



#### Convention

Put each pipe step on its **own line**, with the `|>` at the **end** of the previous line. Indent the continuation by 2 spaces.

Good:

```
penguins |>
  filter(!is.na(body_mass_g)) |>
  group_by(species) |>
  summarize(avg_mass = mean(body_mass_g))
```

Not so good (works, but much harder to read):

```
penguins |> filter(!is.na(body_mass_g)) |> group_by(species) |> summarize(avg_mass = mean(body_mass_g))
```

## Two pipes?

`|>` vs. `%>%`

You will see **two** pipe operators in R code online:

Table 1: The two pipes in R.

| Pipe                | Where it comes from                               | When introduced |
|---------------------|---------------------------------------------------|-----------------|
| <code>%&gt;%</code> | The <code>magrittr</code> package (and tidyverse) | 2014            |
| <code> &gt;</code>  | Base R                                            | 2021            |

They do **almost the same thing**. There are slight differences, but for the purposes of this course, the distinction is not very significant.

### Why two pipes?

- `%>%` came first. It's the original tidyverse pipe from the `magrittr` package.
- Base R added `|>` later because the pipe became so popular they wanted it built in.
- Note that the textbook (*Data Computing*) uses `%>%`; I'll mostly use `|>`.

#### Note

**You can use either one.** But try to be consistent within a single document.

### Typing the pipe quickly

You don't have to type `|>` character-by-character. RStudio has a shortcut:

- **Mac:** `Cmd + Shift + M`
- **Windows/Linux:** `Ctrl + Shift + M`

#### Tip

By default, this shortcut inserts `%>%`. To make it insert `|>` instead:  
Tools > Global Options... > Code > Use native pipe operator

## Reading piped code

### Translating between styles

You'll see code in both styles all over the internet. Practice translating in your head.

Nested:

```
head(arrange(filter(penguins, species == "Gentoo"), body_mass_g), 5)
```

Piped:

```
penguins |>
  filter(species == "Gentoo") |>
  arrange(body_mass_g) |>
  head(5)
```

This gives the same result, and is **much** easier to read the piped version.

### Limitations of the pipe

⚠ The pipe doesn't run things in parallel

It just rewrites your code. Each step still runs one at a time, in order.

⚠ The pipe is NOT the same as + in ggplot2

- dplyr and other data verbs: use |> or %>%.
- ggplot2 layers: use + (it was developed before the pipe).

```
# Correct:
penguins |>
  filter(!is.na(body_mass_g)) |>
  ggplot(aes(x = body_mass_g)) +
  geom_histogram()
```

## A peek at what's coming

You'll see the pipe very often from here on out. Next time we start the **dplyr** verbs:

- `filter()` - pick rows
- `select()` - pick columns
- `arrange()` - sort
- `mutate()` - make new columns
- `summarize()` - collapse to a single number
- `group_by()` - split into groups before summarizing

Every one of these is designed to be chained together with the pipe.

## In-class activity

### Activity: pipe practice (~ 15 min)

- [Click here for the activity](#)

**To turn in:** Save your script. We'll check at the end of class.

### Activity debrief

Common things to watch for:

- **The pipe goes at the end of the line**, not the start. `x |>` then a new line - not `\n |> f()`.
- **The first argument is filled automatically.** Don't write `penguins |> filter(penguins, ...)`: drop the first `penguins`.
- **+ vs. |> again.** `dplyr` verbs chain with `|>`; `ggplot` layers chain with `+`.
- **Parentheses still matter.** `f()` not `f`. The pipe doesn't change that.

### Wrap-up: what we covered today

- The pipe operator (`|>` or `%>%`) takes the value on the left and feeds it as the first argument to the function on the right.
- Pipes let you read code top-to-bottom in the order you'd actually do the work.
- Each step gets its own line; the pipe lives at the **end** of the previous line.
- `|>` and `%>%` are nearly interchangeable for our purposes.
- `dplyr` verbs chain with `|>`; `ggplot` layers chain with `+`.

## Wrap-up & looking ahead

**Tuesday:** Data frames · intro to data graphics.

### Upcoming Deadlines

- Today, 11:59 p.m. - [HW 1: Basic R & Plots](#)
- Mon 5/25, before class - [DC §5.1-5.6](#)

### Reach out

Stuck? Office hours, or email me ([cgc5478@psu.edu](mailto:cgc5478@psu.edu)) or Xinyue ([xpw5228@psu.edu](mailto:xpw5228@psu.edu)).

### Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · *[Data Computing](#)*